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Covariates in DiD

Covariates

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Your DiD lives or dies by parallel trends

With **no anticipation** and a **clean (untreated) comparison group**, the 2×2 is always and only:

$$\widehat{2 \times 2} = \text{ATT} + \text{non-PT bias}$$

- PT is the **only** condition you need for unbiasedness.
- The 2×2 *is whatever it is*. No estimation trick saves it. If PT holds, the number you computed yesterday *is* the ATT.
- So your paper's evidence is mostly about **defending PT** — not about estimation.

What if PT fails?

Then to keep using DiD, you have to *recover* it. Two options:

1. **Triple differences.** A second between-group difference absorbs the differential trend.
   Harder design. Recent audits suggest most papers are doing it incorrectly.
2. **Covariates.** Condition on X so that PT holds *within strata of* X, even if it fails unconditionally.

Today: option 2. Covariates in DiD serve a non-intuitive purpose — they're not for "controlling." They're for fixing broken parallel trends.

Covariates as the fix for broken PT

We'll cover a variety of estimators that incorporate covariates into the DiD estimation:

- **OR** — outcome regression (Heckman, Ichimura & Todd 1997)
- **IPW** — inverse propensity weighting (Abadie 2005)
- **DR-DiD** — doubly robust (Sant'Anna & Zhao 2020)

We'll contrast each with the workhorse: **OLS TWFE with controls**. *Spoiler: TWFE-with-controls is the false friend. It looks like it should work. It doesn't.*

The applied problem

- Your treated states differ from your controls on observables — baseline income, demographics, industry mix, education.
- If $Y(0)$ trends differ across these dimensions, unconditional PT is implausible.
- But maybe *within* strata of X, PT holds:

$$\mathbb{E}[\Delta Y(0) \mid D = 1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D = 0, X]$$

- This is **conditional parallel trends**. The escape valve for designs without a placebo group.
- Three estimators target this. They differ in *which side* of the conditioning they model.

This afternoon, six beats

1. Outcome regression (Heckman-Ichimura-Todd, 1997) — the bias OLS-with-controls actually identifies, plus the conditional-PT assumption that anchors the rest of the afternoon
2. Inverse propensity weighting (Abadie 2005)
3. Doubly robust DiD (Sant'Anna & Zhao 2020)
4. Hidden linearity bias — TWFE-with-controls fully exposed (Caetano & Callaway 2024)
5. Choosing & checking X — the practical close

We lead with the bias of OLS. Each fix below removes one piece of that bias.

Lesson

1. Outcome Regression

What OLS-with-controls gives you — and what it doesn't

Conditional parallel trends, formally

The new identifying assumption

$$\mathbb{E}[\Delta Y(0) \mid D{=}1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$$

- Within strata of X, the untreated trend is the same for treated and control.
- Equivalently: *any* differential trend across treated and control is fully explained by their differences in X.
- Trade-off: weaker than unconditional PT (good), but you have to pick the right X (hard).
- Two ways to use X: model the outcome side ($\mu(X)$), or model the treatment side ($p(X)$). Each is one of our estimators.

A note on history of thought

The first articulation of any parallel-trends-type assumption in the DiD literature is:

Heckman, Ichimura & Todd (1997), ***Review of Economic Studies***

“Matching as an econometric evaluation estimator: evidence from evaluating a job training programme.”

The irony. What HIT wrote down was the ***conditional*** PT assumption — $\mathbb{E}[\Delta Y(0) \mid D{=}1, X] = \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$.

The TWFE-with-controls regression — and the three assumptions it hides

$$Y_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \delta\,(D_i \times \text{post}_t) \;+\; \beta\, X_{it} \;+\; \varepsilon_{it}$$

"Sure! If you're willing to impose three more assumptions."

$$\underbrace{\delta \text{ doesn't depend on } X}_{\textbf{Bias}_{\text{HTE}}} \quad \underbrace{X \text{ has parallel trends across groups}}_{\textbf{Bias}_{\Delta X}} \quad \underbrace{X \text{ is time-invariant}}_{\textbf{Bias}_{\text{FE}}}$$

- **Bias$_{\text{HTE}}$**. If $\delta(X)$ varies with X, TWFE fits one δ. *Y(1) problem. Sign indeterminate.*
- **Bias$_{\Delta X}$**. If X is imbalanced and drives $Y(0)$'s trend, PTA fails. *Y(0) problem. Sign indeterminate.*
- **Bias$_{\text{FE}}$**. α_i demeans time-invariant X to zero, silently restoring *unconditional* PT. Fix: Post $\times$ $X_{i,0}$. *Sign indeterminate.*

Proof, step 1 — the TWFE model and what conditional PT gives us

The TWFE model with one covariate X:

$$Y_{it} \;=\; \alpha_1 + \alpha_2\, T_t + \alpha_3\, D_i + \delta\,(T_t \times D_i) + \beta\, X_{it} + \varepsilon_{it}.$$

Conditional parallel trends says, for each value of X:

$$\mathbb{E}[Y_1^0 - Y_0^0 \mid D{=}1, X] \;=\; \mathbb{E}[Y_1^0 - Y_0^0 \mid D{=}0, X].$$

Rearrange. The missing counterfactual for the treated is identified:

$$\mathbb{E}[Y_1^0 \mid D{=}1, X] \;=\; \mathbb{E}[Y_0 \mid D{=}1, X] + \mathbb{E}[Y_1 \mid D{=}0, X] - \mathbb{E}[Y_0 \mid D{=}0, X].$$

Three observed conditional means $\Rightarrow$ one unobserved counterfactual. That's all PT does for us.

Proof, step 2 — plug TWFE in, the slopes look the same

From the TWFE model, the observed conditional means are:

$$\mathbb{E}[Y_0 \mid D{=}1, X] = \alpha_1 + \alpha_3 + \beta\, X$$
$$\mathbb{E}[Y_1 \mid D{=}0, X] = \alpha_1 + \alpha_2 + \beta\, X$$
$$\mathbb{E}[Y_0 \mid D{=}0, X] = \alpha_1 + \beta\, X$$

Plug into the identity from step 1:

$$\mathbb{E}[Y_1^0 \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \beta\, X.$$

The treated Y^1 post-period, from the TWFE model:

$$\mathbb{E}[Y_1^1 \mid D{=}1, X] \;=\; \alpha_1 + \alpha_2 + \alpha_3 + \delta + \beta\, X.$$

Both potential outcomes carry the same β in X. Subtract: ATT $= \delta$. Clean. Suspiciously clean.

Proof, step 3 — what if Y^1 and Y^0 have different slopes in X?

Let the slopes differ. Write β_1 for the Y^1 slope and β_2 for the Y^0 slope:

$$\begin{aligned}
\mathbb{E}[Y_1^1 \mid D{=}1, X] &= \alpha_1 + \alpha_2 + \alpha_3 + \delta + \boxed{\beta_1}\, X \\
\mathbb{E}[Y_1^0 \mid D{=}1, X] &= \alpha_1 + \alpha_2 + \alpha_3 \qquad\quad + \boxed{\beta_2}\, X
\end{aligned}$$

The true ATT given X:

$$\text{ATT}(X) \;=\; \mathbb{E}[Y^1 - Y^0 \mid D{=}1, X] \;=\; \delta \;+\; (\beta_1 - \beta_2)\, X.$$

- If $\beta_1 \neq \beta_2$, the treatment effect *depends on* X.
- TWFE fits **one** β, not two.
- So $\widehat{\delta}$ recovers the ATT only when $\beta_1 = \beta_2$.

Build $Y(0)$ and $Y(1)$ by hand — homogeneous vs. heterogeneous in X

Same $Y(0)$ in both. The treatment effect $Y(1) - Y(0)$ is either a *constant* or a *function of* X.

Homogeneous TE in X (Assumption 4 holds)

```
gen y0 = 15000 + 10.25*age - 10.5*age_sq ///
             + 1000*gpa - 10.5*gpa_sq ///
             + 500*age*gpa + e
label var y0 "Earnings w/o degree"

gen y1 = y0 + 2500
label var y1 "Earnings with degree"

* delta = 2500 for everyone.
```

Heterogeneous TE in X (Assumption 4 fails)

```
gen y0 = 15000 + 10.25*age - 10.5*age_sq ///
             + 1000*gpa - 10.5*gpa_sq ///
             + 500*age*gpa + e
label var y0 "Earnings w/o degree"

gen y1 = y0 + 2500 + 100*age + 1100*gpa
label var y1 "Earnings with degree"

* delta(X) = 2500 + 100*age + 1100*gpa.
```

- **Left:** $Y(0)$ and $Y(1)$ share *the same* slope on X. ATT $= 2{,}500$ for everyone. TWFE-with-X recovers it.
- **Right:** $Y(1)$ slopes differ. A 1-year-older worker gains \$100 more; a 1-GPA-point higher worker gains \$1,100 more. **TWFE fits one δ, misses the variation.**

Proof, step 4 — Assumption 4: homogeneous δ in X

Assumption 4 (Homogeneous treatment effects in X)

TWFE-with-X requires the treatment effect to be the same at every value of X.

- If X is **sex**: the effect is the same for men and for women.
- If X is **income** (continuous): the effect is the same at \$1 and at \$1,000,000.
- If X is **age**: the effect is the same at 25 and at 65.

This is a structural assumption on the DGP. *OLS doesn't tell you you're making it. The math does.*

If the effect actually varies with X, TWFE-with-X averages across the variation and gives you a weighted parameter you didn't ask for.

Bias$_{\Delta X}$: the RCT analogy

Earnings Y depends on a million Xs: gender, age, education, neighborhood, race, occupation, parental income, ...

Randomization *balances* those Xs across treated and control.

The $X \to Y$ channels cancel.

DiD doesn't get balance for free

If treated and control look different on X, and those Xs drive how $Y(0)$ moves over time, the group-level $Y(0)$ trends diverge.

And PTA on $Y(0)$ fails.

A simple example: gender

$\Delta Y(0)$ over the panel:

- Men: $+\$2$
- Women: $+\$1$

No treatment effect anywhere. Earnings just drift up faster for men than for women.

Composition: **75% male in treated, 25% male in control.**

Naive DiD reads the gap

$$\mathbb{E}[\Delta Y(0) \mid D{=}1] \;=\; 0.75(2) + 0.25(1) \;=\; 1.75$$
$$\mathbb{E}[\Delta Y(0) \mid D{=}0] \;=\; 0.25(2) + 0.75(1) \;=\; 1.25$$

Naive DiD: $1.75 - 1.25 = +0.50$ **— pure imbalance bias.**

Formal: Y(0) trend depends on X

Let

$$Y(0)_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \beta_t\, X_i \;+\; \varepsilon_{it}$$

where X_i is time-invariant but its coefficient β_t can drift.

Differencing pre-to-post:

$$\Delta Y(0)_i \;=\; \Delta\gamma \;+\; (\beta_{\text{post}} - \beta_{\text{pre}}) \cdot X_i.$$

The bias factors into trend $\times$ imbalance

$$\text{Bias}_{\Delta X} \;=\; \underbrace{(\beta_{\text{post}} - \beta_{\text{pre}})}_{X\text{-specific trend in } Y(0)} \;\times\; \underbrace{\left(\mathbb{E}[X \mid D{=}1] - \mathbb{E}[X \mid D{=}0]\right)}_{\text{imbalance on } X}$$

It's a product. Either factor at zero kills it.

Either factor at zero kills it

- Balance X across groups $\Rightarrow$ no bias, even with strong β_t drift.
- Constant β $\Rightarrow$ no bias, even with strong imbalance.

Imbalance is the root

Randomization (in an RCT) zeroes the imbalance term — so it doesn't matter what X-specific dynamics exist.

Without randomization, conditioning on X does the same job.

DR / IPW / OR are just different ways to re-weight toward balance.

The gender example in Stata

```
* Gender is time-invariant -- but its effect on Y(0) drifts over time.
* delta Y(0) = +2 for men, +1 for women (no treatment effect)
* Composition: 75% male in treated, 25% male in control.

gen y0_pre = 1000 + 100*male + rnormal(0, 5)
gen y0_post = y0_pre + 2*male + 1*(1 - male)

mean y0_post if treat == 1 // 1.75 in expectation
mean y0_post if treat == 0 // 1.25 in expectation
```

$1.75 - 1.25 = 0.50$. **Imbalance $\times$ drift in β_t.**

A worked example of the violation

We'll build a world where:

- the true treatment effect is *positive*,
- the covariate's effect on $Y(0)$ *changes sharply at the year of treatment*,
- and the covariate is *imbalanced* across treated and control.

Then we run a plain DiD and see what we get.

What we built in

500 counties, 10 years, treatment in year 6. Outcome: county birth rate.

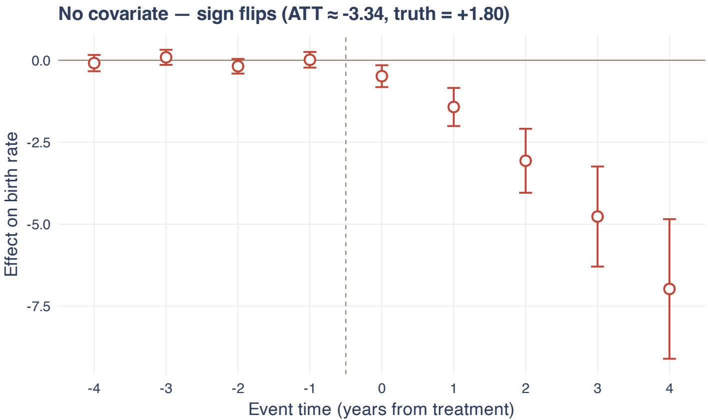
- **True effect:** $+0.6$ at year 6 $\to +3.0$ by year 10. ATT $\approx +1.8$.
- **Covariate trend:** urban $Y(0)$ slope shifts from ~ -0.1 to -2.5 at year 6. Rural flat.
- **Imbalance:** $\Pr(\text{treated} \mid \text{urban}) = 0.70$, $\Pr(\text{treated} \mid \text{rural}) = 0.30$.

Changing β_t on urban, kicking in exactly when treatment does.

The DGP

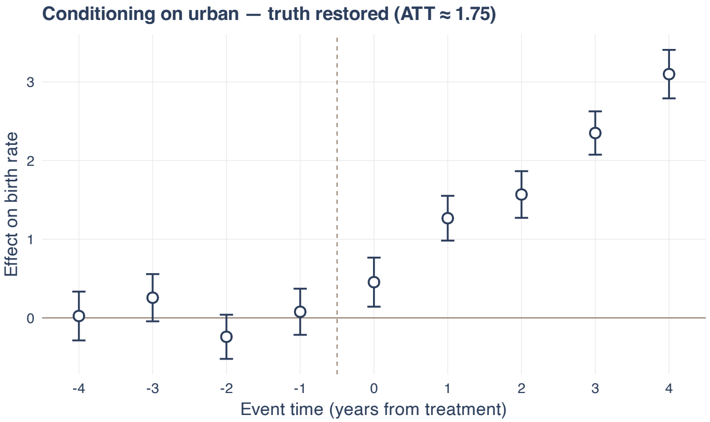
```
set seed 12345
set obs 500
gen county_id = _n
gen urban = (runiform() < 0.5)
gen treated = (urban==1 & runiform()<0.7) | (urban==0 & runiform()<0.3)
gen treat_date = cond(treated, 6, 0)
expand 10
bysort county_id: gen year = _n
gen post = (year >= 6)

* Smaller urban decline, nearly flat rural increase
bys county_id: gen trend = .
replace trend = -0.15*urban + 0.1*(1-urban) if year==1
replace trend = -0.25*urban + 0.1*(1-urban) if year==2
replace trend = -0.15*urban + 0.1*(1-urban) if year==3
replace trend = -0.10*urban + 0.1*(1-urban) if year==4
replace trend = -0.10*urban + 0.1*(1-urban) if year==5
replace trend = -0.50*urban + 0.1*(1-urban) if year==6
replace trend = -1.00*urban + 0.1*(1-urban) if year==7
replace trend = -1.50*urban + 0.1*(1-urban) if year==8
replace trend = -2.00*urban + 0.1*(1-urban) if year==9
replace trend = -2.50*urban + 0.1*(1-urban) if year==10

bys county_id: gen county_fe = rnormal(0, 2)
bys county_id: gen u = rnormal(0, 1)
gen y0 = 15 + county_fe + trend*year + u
gen y1 = y0 + post*treated*(year - 5)*0.6
gen birth_rate = treated*y1 + (1-treated)*y0
```

Run a DiD event study



Pre-trends flat. Truth was $+1.8$. We got -3.3.

We can recover this with the methods today



Same event study, conditioned on urban. ATT $\approx +1.75$.

One extra variable

```
* Naive
estat event after a plain DiD // ATT ~ -3.3

* Conditioned on urban
estat event after a DiD with urban // ATT ~ +1.75
```

We'll meet the estimators that do this — DR, IPW, OR — next.

Pre-trends don't protect us from imbalance

The pre-trends test asks: *did $Y(0)$ move in parallel before treatment?*

It can't see a β_t-on-X shock that turns on at the year of treatment — the one that imbalance is about to amplify.

Pre-trends are necessary. They are not sufficient.

Balance on X is the second piece of evidence we need.

Heuristic check: balance tables on X

You can't test PTA on $Y(0)$ directly — it's counterfactual.

What you CAN check: is X balanced across treated and control?

- **Levels** ($\mathbb{E}[X \mid D{=}1] = \mathbb{E}[X \mid D{=}0]$?) — matters when X's effect varies over time.
- **Trends** ($\mathbb{E}[\Delta X \mid D{=}1] = \mathbb{E}[\Delta X \mid D{=}0]$?) — matters under constant β.

Baker et al. (2025, *Journal of Economic Literature*) Table 2 reports both: a panel of pre-period X levels and a panel of X differences across years. Both panels matter because both are channels through which imbalance creates $\text{Bias}_{\Delta X}$.

Identification vs. heuristic. PTA on $Y(0)$ is the identifying assumption. Balance on X is a heuristic check. Same epistemic status as the pre-period parallel-trends test: it's evidence, not proof.

Time-varying X: the bad-control problem

The trap. A time-varying covariate is measured both before *and after* treatment. If $D \to X$ (treatment changes X), then post-period X is not a pre-treatment characteristic — it's a partial *outcome*.

- Conditioning on post-treatment X blocks part of the causal channel $D \to Y$ (mediator bias).
- Worse, if X has unobserved common causes with Y, conditioning opens a back-door collider path (Angrist–Pischke "bad control").

The condition you need. $X(1) = X(0)$ — treatment doesn't move X. Equivalently: use only pre-period X, since by no-anticipation $X_{i,0} = X_i(0)$ mechanically. *Rule of thumb: don't condition on anything that could be a downstream consequence of D. Demographics and baseline X are safe. Income or employment status measured post-treatment is not.*

Time-invariant baseline X: eaten by the fixed effect

The other trap. You wisely use *baseline* (pre-period) X to avoid the bad-control problem above. But then:

- Within-unit demeaning makes $\tilde{X}_{it} = X_{it} - \bar{X}_i = 0$ if X is time-invariant.
- So `race`, `education at baseline`, `pre-program earnings` — the kind of X you needed to defend CPT — vanish.
- Additive-X TWFE estimates as if you'd never included them.

The fix. Enter X as Post $\times$ $X_{i,0}$. Post varies within unit, so the FE cannot eat the interaction. *The double bind: time-varying X is a bad control; time-invariant X gets eaten by α_i.*

Post $\times$ $X_{i,0}$ is the escape valve.

Proof, step 5 — the goal: parallel trends on Y^0, and TWFE's model

What we need. Parallel trends on the untreated potential outcome:

$$\mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1] \;=\; \mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}0]$$

TWFE's structural model for Y^0:

$$Y^0_{it} \;=\; \alpha_1 \;+\; \alpha_2\, T_t \;+\; \alpha_3\, D_i \;+\; \beta\, X_{it} \;+\; \varepsilon^0_{it}$$

β is the slope of Y^0 on X. The question for the next slides: what does the model assume to make PT hold?

Proof, step 6 — Y^0's within-group trend, by subtraction

For group D, write the post and pre conditional means:

$$\begin{aligned}\mathbb{E}[Y^0_{\text{post}} \mid D] &= \alpha_1 + \alpha_2 + \alpha_3 D + \beta\,\mathbb{E}[X \mid D, \text{post}] \\ \mathbb{E}[Y^0_{\text{pre}} \mid D] &= \alpha_1 \qquad\quad + \alpha_3 D + \beta\,\mathbb{E}[X \mid D, \text{pre}]\end{aligned}$$

Subtract: the α_1 and $\alpha_3 D$ cancel.

$$\boxed{\mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D] = \alpha_2 + \beta \cdot \Delta X^D}$$

where $\Delta X^D \equiv \mathbb{E}[X|D, \text{post}] - \mathbb{E}[X|D, \text{pre}]$.

Proof, step 7 — PT forces $\beta\,(\Delta X^T - \Delta X^C) = 0$

Set the two trends equal (PT on Y^0):

$$\alpha_2 + \beta\,\Delta X^T \;=\; \alpha_2 + \beta\,\Delta X^C$$

The α_2 cancels. Rearrange:

$$\boxed{\beta\,(\Delta X^T - \Delta X^C) \;=\; 0}$$

Two ways for this to hold:

- $\beta = 0 \;\Rightarrow\; Y^0$ doesn't depend on X.
- $\Delta X^T = \Delta X^C \;\Rightarrow\; X$-trends parallel across groups.

Proof, step 8 — Assumptions 5 & 6: Y^0 trends don't depend on X

Assumptions 5 & 6

Y^0's trend over time does not depend on X — within either the treated or the control group.

- Rich people cannot trend differently than poor people.
- Men cannot trend differently than women.
- College grads cannot trend differently than high-school dropouts.

The realized-X version ("X-trends equal") is a consequence, not the assumption.

Without these six (1–3 baseline DiD plus 4, 5, 6), TWFE-with-X does not identify the ATT.

Closer: from the model to the data via the switching equation

The proof worked entirely in $Y(0)$ — a counterfactual we don't directly observe.

The switching equation tells us when Y *is* $Y(0)$:

$$Y_{it} \;=\; D_i\,\text{post}_t \cdot Y_{it}(1) \;+\; (1 - D_i\,\text{post}_t) \cdot Y_{it}(0).$$

- In three of the four 2×2 cells, $Y = Y(0)$. The TWFE-with-X model fits those directly.
- In the treated-post cell, $Y = Y(1)$. The model gives us $E[Y(0) \mid D{=}1, \text{post}, X]$ via Assumptions 5–6 — the imputed counterfactual.
- Subtract the imputed $Y(0)$ from the observed $Y(1)$. That difference, averaged over the treated X-distribution, is the ATT.

The algebra is in $Y(0)$. The switching equation is what makes the algebra a statement about observable data.

Four ways to break TWFE-with-X

Generate $Y(0)$ *directly*, cell-by-cell — so the true DGP isn't $\alpha + \delta D + \beta X + \varepsilon$. Each version turns on one knob.

- **DGP 0 — clean linear.** $Y(0) = \alpha_i + \gamma_t + \beta X + \varepsilon$. No violation. Sanity check.
- **DGP 1 — time-varying β.** $\beta_{\text{pre}} = 1,\ \beta_{\text{post}} = 3$. X's effect on Y shifts at the policy break.
- **DGP 2 — heterogeneous δ in X.** $\delta_i = 2 + 4(X_i - \frac{1}{2})$. Treatment effect grows with X.
- **DGP 3 — time-invariant Z.** $Y(0)$ depends on a baseline Z only in post. FE within-unit transformation *eats* Z.

True $\delta = 2$ in every DGP. Selection: treated units' X shifts up by 0.3 in post (the typical empirical pattern).

The DGP, in R

```r
# Common setup
N <- 4000
units <- data.frame(id = 1:N,
                    D = rbinom(N, 1, 0.5),
                    Z = rbinom(N, 1, 0.5))
panel <- merge(units, data.frame(t = c(0, 1)))
panel$post <- as.integer(panel$t == 1)
panel$X <- runif(nrow(panel)) + 0.3 * panel$D * panel$post

# Y(0), one of four versions:
panel$Y0 <- panel$alpha + 0.5*panel$post + 1*panel$X + panel$eps # DGP 0
# Y0 = ... + ifelse(panel$post == 0, 1, 3) * panel$X + ... # DGP 1
# tau = 2 + 4*(X - 0.5) # DGP 2
# Y0 = ... + 5*panel$Z*panel$post + ... # DGP 3

panel$Y <- panel$Y0 + panel$D * panel$post * panel$tau
```

Full script: `decks/code/Covariates-Bias/twfe_bias_dgp.R`.

Results: which spec fixes which bias? (true $\delta = 2$)

	no covariates	additive X	Post$\times X$	Post$\times X + Z$
DGP 0 clean	2.30	2.02	2.01	2.01
DGP 1 time-varying β	2.91	2.30	1.99	1.99
DGP 2 het δ in X	3.51	2.91	2.64	2.64
DGP 3 time-inv Z	2.15	1.85	1.83	1.98

green : spec recovers $\delta \approx 2$. *red : spec is biased.*

The bias-to-fix map

- **Time-varying** β $\rightarrow$ the fix is Post$\times X$. Let X's effect move with the policy break.
- **Heterogeneous** δ **in** X $\rightarrow$ Post$\times X$ *still misses it.* You need a saturated $D\times X$ interaction, or DR-DiD (Lesson 4), or matching.
- **Time-invariant** Z $\rightarrow$ the fix is Post$\times Z$. Additive Z does nothing because FE absorbs it.

The unifying lesson (Caetano & Callaway 2024)

Additive-X TWFE is one regression that silently assumes *all three* of these don't apply. When they do, the bias has a name: **hidden linearity bias.** Each fix is a relaxation.

The proper TWFE spec — and what it still can't handle

To kill Bias$_{\Delta X}$ and Bias$_{\text{FE}}$ inside TWFE, you need:

$$Y_{it} \;=\; \alpha_i + \gamma_t + \delta\,(D_i \times \text{post}_t) \;+\; X'_{i,0}\gamma_X \;+\; (\text{post}_t \cdot X_{i,0})'\gamma_{pX} \;+\; (\text{post}_t \cdot Z_i)'\gamma_{pZ} \;+\; \varepsilon_{it}$$

- $(\text{post} \times X_{i,0})$ handles Bias$_{\Delta X}$: lets the *pre-period* value of X get a different slope in post.
- $(\text{post} \times Z)$ handles Bias$_{\text{FE}}$: rescues time-invariant baseline characteristics from the FE.

Note the subscript $i, 0$. Use the *pre-period* value of X throughout. If you interact D or post with the contemporaneous X_{it}, you're conditioning on a post-treatment variable — bad control (Angrist–Pischke). Baseline $X_{i,0}$ satisfies $X_{i,0}(1) = X_{i,0}(0)$ by construction.

What this spec *cannot* fix: Bias$_{\text{HTE}}$. If δ depends on X, no single $\hat{\delta}$ recovers the ATT.

Saturated TWFE *on* $X_{i,0}$ mirrors what OR/IPW/DR do

OR, IPW, and DR condition on *baseline* $X_{i,0}$ only — never X_{it}. A deliberate choice to avoid post-treatment bad control. An honest TWFE comparison has to make the same commitment.

Saturated TWFE on $X_{i,0}$:

$$Y \;=\; \alpha_i + \gamma_t + (D\times\text{post})\left[\delta_0 + X_{i,0}'\delta_X\right] + X_{i,0}'\gamma + (\text{post}\cdot X_{i,0})'\gamma_p + \cdots$$

Every baseline covariate gets $D\times X_{i,0}$, post$\times X_{i,0}$, and $D\times$post$\times X_{i,0}$ interactions. Then average $\hat{\delta}(X_{i,0})$ over the treated $X_{i,0}$-distribution.

In principle it works; in practice it's a minefield:

- **True saturation discretizes X.** Continuous X has to be binned. Rare cells $\rightarrow$ sparseness.
- **Swap $X_{i,0}$ for X_{it} and the equivalence breaks** — post-treatment bad control.
- **Headline coefficient at $X = 0$ isn't the ATT.** Average $\hat{\delta}(X_{i,0})$ over the treated.

OR, IPW, DR commit to baseline X structurally. The estimator does saturation *and* the ATT average for you. *(Time-varying X is a separate problem they don't claim to solve.)*

Why this matters in practice

The methodology isn't just elegance — it's protection. Three traps researchers actually fall into:

- **Post-treatment X used as a control.** Bad control. Almost no one explicitly defends pre-period X, but the choice changes the answer.
- **Software defaults.** `xtreg`, `reghdfe` silently drop a base category. The dropped category is the implicit reference point; coefficient interpretation depends on which one was dropped. A misread of the printed table can flip a sign.
- **Reporting the coefficient as "the ATT"** when it's the effect *at* $X = 0$ — not the ATT averaged over the treated X-distribution.

OR / IPW / DR don't make these mistakes available. The saturation lives inside the estimator. Defaults are sensible. The output is the ATT.

The point isn't that saturated TWFE is wrong. It's that it's a knife held the wrong way. OR/IPW/DR are the same knife, sheathed.

Three doors to the same fix

Three estimators that handle heterogeneous $\delta(X)$ — each does it differently:

- **Outcome regression (OR), Heckman, Ichimura & Todd (1997).**
  Fit $\widehat{\mu}_0(X)$ on controls. Impute the counterfactual for each treated unit. Average over the treated X-distribution. *This lesson.*
- **Inverse propensity weighting (IPW), Abadie (2005).**
  Reweight the controls by $\widehat{p}(X)/(1-\widehat{p}(X))$ to match the treated X-distribution. *Lesson 2.*
- **Doubly-robust DiD, Sant'Anna & Zhao (2020).**
  Combine OR and IPW. Consistent if *either* model is right. *Lesson 3.*

Three doors, same room. The choice between them is mostly a choice about which model you'd rather defend.

A long lineage: Oaxaca, Blinder, and what's coming

The wage-gap question (1973). Two groups have different average wages. How much of the gap is differences in X (education, experience)? How much is differences in *returns* to X?

Oaxaca (*IER* 1973) and Blinder (*JHR* 1973) fit a separate regression in each group and asked:

What would Group A's wages look like under Group B's coefficients?

The same idea travels:

- **Heckman–Ichimura–Todd (1997).** Import into DiD as *outcome regression*.
- **Wooldridge & Słoczyński (recent).** TWFE with rich $D \times X$ interactions *is* regression adjustment.
- **Borusyak–Jaravel–Spiess (2024).** "Imputation estimator" — numerically identical to TWFE-RA in staggered DiD.

Different vocabularies. Same machinery for fifty years. HIT 1997 is the DiD form. Here's how it works.

First-difference: DiD becomes cross-sectional Oaxaca-Blinder

The step that does the heavy lifting. Take $\Delta Y_i = Y_{i,\text{post}} - Y_{i,\text{pre}}$. Each unit's two-period panel collapses to *one number*.

The DiD problem becomes a cross-section:

compare ΔY across $D = 0$ and $D = 1$, conditional on X.

This is Oaxaca-Blinder's home turf. Two groups, one outcome (now ΔY), a covariate vector, the same counterfactual question — evaluate one group's data at the other group's regression coefficients.

Bonus, structural. The X in the cross-section is mechanically the *baseline* X — measured at $t = 0$, before treatment. Post-treatment X never enters the regression.

OR has no bad-control problem by construction. FD solved it before we asked.

Same machine, three targets: ATE, ATT, ATU

The cross-sectional regressions on ΔY give you a conditional treatment effect at each X:

$$\delta(X) \;=\; \widehat{\mu}_1(X) - \widehat{\mu}_0(X) \;=\; \widehat{\mathbb{E}}[\Delta Y \mid D{=}1, X] - \widehat{\mathbb{E}}[\Delta Y \mid D{=}0, X]$$

What you average $\delta(X)$ over determines the estimand:

Average over	Estimand	Question it answers
all units, full X distribution	ATE	"population-average effect"
treated units, $X \mid D{=}1$	**ATT**	"effect on those who took it"
control units, $X \mid D{=}0$	ATU	"effect on those who didn't"

DiD's natural target is the ATT. Random assignment is broken; treatment is selected; the policy question is the effect on the population that actually took it.

So OR averages $\delta(X)$ over the *treated* X-distribution. Same model. Choose the average. Get the estimand.

Recipe A: impute the counterfactual

Heckman, Ichimura & Todd (1997).

1. Use *only the control group* to fit a model of ΔY on X:

$$\widehat{\mu}_0(X) \;=\; \widehat{\mathbb{E}}[\Delta Y \mid D{=}0, X]$$

2. For each treated unit, predict the counterfactual trend $\widehat{\mu}_0(X_i)$.
3. The OR-DiD estimator:

$$\widehat{\delta}_{\text{OR}} \;=\; \frac{1}{N_1} \sum_{i:\, D_i=1} \big(\Delta Y_i \;-\; \widehat{\mu}_0(X_i)\big)$$

Interpretation: OR *imputes* what the treated would have done if untreated — using only the control group's X-to-trend mapping.

Recipe B: the same OR, written as one regression

Instead of fitting on controls and predicting, write *one* saturated regression on the full sample:

$$\Delta Y_i \;=\; \alpha \;+\; \beta\, X_i \;+\; \gamma\, D_i \;+\; \delta\,(D_i \cdot X_i) \;+\; \varepsilon_i$$

Then read off the ATT:

- Marginal effect of D at $X = X_i$: $\quad \gamma + \delta\, X_i$
- Average over the treated X-distribution: $\quad \widehat{\delta}_{\text{sat}} \;=\; \gamma + \delta \cdot \overline{X}_{D=1}$

Both recipes ask the same question: what would treated units' ΔY look like if they had controls' X-to-trend mapping?

Recipe A imputes the answer one unit at a time. Recipe B fits everyone jointly and reads it off. Same answer either way — but only if the regression is properly saturated.

Same answer, different machinery?

Recipe A (HIT 1997) uses *observed* ΔY on the treated:

$$\widehat{\delta}_A \;=\; \frac{1}{N_1}\sum_{D=1}\Big[\Delta Y_i \;-\; \widehat{\mu}_0(X_i)\Big]$$

Recipe B (saturated regression) uses *fitted* $\widehat{\mu}_1(X)$ on the treated:

$$\widehat{\delta}_B \;=\; \frac{1}{N_1}\sum_{D=1}\Big[\widehat{\mu}_1(X_i) \;-\; \widehat{\mu}_0(X_i)\Big]$$

The first term is different. One is raw data, the other is a model prediction.

And yet, on LaLonde, both gave \$1,770 to the dollar. Let's see why.

One OLS condition does the work

The saturated regression chooses $(\widehat{\alpha}, \widehat{\beta}, \widehat{\gamma}, \widehat{\delta})$ to minimize the sum of squared residuals.

Each coefficient gets a first-order condition. The one we need is for $\widehat{\gamma}$ (the coefficient on D):

$$\sum_i D_i \cdot \varepsilon_i \ = \ 0$$

In words. Sum the residuals from treated observations only. OLS forces that sum to zero.

The treated residuals net out to zero. Exactly, by construction.

Plug in what a residual is

For a treated unit i, the fitted value is

$$\widehat{\mu}_1(X_i) \;=\; \widehat{\alpha} + \widehat{\beta} X_i + \widehat{\gamma} + \widehat{\delta} X_i$$

And the residual is *observed minus fitted:*

$$\varepsilon_i \;=\; \Delta Y_i \;-\; \widehat{\mu}_1(X_i)$$

Substitute into the FOC:

$$\sum_{D=1} \Big[\Delta Y_i \;-\; \widehat{\mu}_1(X_i) \Big] \;=\; 0$$

Rearrange

$$\sum_{D=1} \Delta Y_i = \sum_{D=1} \widehat{\mu}_1(X_i)$$

Divide both sides by N_1, the number of treated units:

$$\boxed{\frac{1}{N_1} \sum_{D=1} \Delta Y_i = \frac{1}{N_1} \sum_{D=1} \widehat{\mu}_1(X_i)}$$

Mean of observed ΔY on the treated = mean of fitted $\widehat{\mu}_1$ on the treated.

Unit-by-unit they differ. In the mean they're equal.

The intuition

OLS regression *passes through* the group means.

Whenever the regression includes a D dummy, OLS guarantees the fitted-value mean on the treated subsample matches the observed mean on the treated subsample.

Fitting smooths the variation *around* the mean.

It cannot change the mean itself.

Fitted values and observed values dance around the same average. Average them and you get the same number.

Both recipes give the same number

Recipe B (saturated):

$$\widehat{\delta}_B = \underbrace{\overline{\widehat{\mu}_1}_{D=1}}_{= \overline{\Delta Y}_{D=1} \text{ by FOC}} - \overline{\widehat{\mu}_0}_{D=1}$$

Recipe A (HIT 1997):

$$\widehat{\delta}_A = \overline{\Delta Y}_{D=1} - \overline{\widehat{\mu}_0}_{D=1}$$

First terms equal by the FOC. Second terms equal because the saturated $D \times X$ interactions pin $\widehat{\mu}_0$ to the same controls-only fit Recipe A uses.

$\widehat{\delta}_A = \widehat{\delta}_B$. **LaLonde: \$1,770 = \$1,770.**

Where this equivalence falls apart

The whole argument rests on *one* property of OLS — the FOC that forces treated residuals to sum to zero.

For nonlinear estimators (logit, probit, GLM, ML):

- The treated fitted values do *not* have to average back to the observed treated mean.
- Recipe A (use observed ΔY) and Recipe B (use fitted $\widehat{\mu}_1$) genuinely diverge.
- The two-regression KOB form becomes the principled choice.

Linear OR-DiD is the lucky special case where the choice doesn't matter. That's why HIT 1997's one-sided imputation and saturated TWFE give the same number on LaLonde — and why the equivalence shouldn't be taken for granted in general.

The same estimator, four names

Tradition	Name
Labor economics, 1973	Oaxaca-Blinder decomposition
DiD, 1997	Heckman-Ichimura-Todd outcome regression
TWFE with $D \times X$ interactions	Wooldridge / Słoczyński regression adjustment
Staggered DiD, 2024	Borusyak-Jaravel-Spiess imputation

Four names. One estimator.

When you hear “imputation” on Day 4 (staggered DiD), you’ll already know what it is. You’ve been doing it all day.

Why OR can fail: it's all model

- OR's entire identification rests on $\widehat{\mu}_0(X)$ being correctly specified.
- Get the functional form wrong — a missing nonlinearity, a missing interaction — and OR is biased.
- Especially fragile in the post-period, where you're *extrapolating* from the control distribution of X to the treated distribution.
- If treated units have X-values rare among controls, OR is leaning on the model where the data are thin.

This is why IPW exists — as a different way to use X.

LESSON

2. Inverse Propensity Weighting

The propensity score, in one minute

- Definition: $p(X) = \Pr(D{=}1 \mid X)$ — the probability of treatment given covariates.
- Estimated by logit/probit **maximum likelihood**:

$$\widehat{\theta} = \arg\max_{\theta} \sum_i \left[\, D_i \log p(X_i;\theta) + (1 - D_i) \log(1 - p(X_i;\theta)) \,\right]$$

- **Rosenbaum-Rubin (1983)**: if D is unconfounded given X (a vector), it is unconfounded given $p(X)$ (a scalar).
  One number summarizes all of X.
- The pscore lives in $[0, 1]$. Treated units have higher $p(X)$ on average. The pscore reorganizes the data along the dimension of selection.

The Rosenbaum-Rubin theorem (1983)

Statement

If treatment is unconfounded given X, i.e. $\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid X$,
then it is also unconfounded given the propensity score $p(X)$ alone:

$$\{Y(0), Y(1)\} \perp\!\!\!\perp D \mid p(X).$$

- X is a vector. $p(X)$ is a scalar in $[0, 1]$. **One number compresses all the confounding information.**
- Operationally: matching, weighting, or balancing on $p(X)$ gives the same identification as on the full X.
- This is what licenses IPW. It does *not* license putting $p(X)$ on the right-hand side of a regression — that's a different use, with different consequences (Lesson 4 trap).

Abadie (2005): reweight the controls

- Idea: reweight control units by $\dfrac{p(X)}{1-p(X)}$ so the reweighted control sample *looks like* the treated sample in X.
- Then take a simple DiD on the reweighted sample.

The Abadie IPW-DiD estimator

$$\widehat{\delta}_{\text{IPW}} = \frac{1}{N_1}\sum_i \left(\frac{D_i - p(X_i)}{1-p(X_i)}\right)\Delta Y_i$$

The operator $(D-p)/(1-p)$ gives weight $+1$ to treated units and weight $-p/(1-p)$ to controls — exactly the reweighting in line one. **Identification:** under conditional PT + common support, IPW recovers the ATT. Proof next slide.

Substitute $D=1$ and $D=0$: the operator does different things to each

The operator we care about:

$$\frac{D - p(X)}{1 - p(X)} \cdot \Delta Y$$

For a $D=1$ row:

$$\frac{1 - p(X)}{1 - p(X)} \cdot \Delta Y \;=\; \Delta Y$$

Just the treated unit's change. **No reweighting.** **For a $D=0$ row:**

$$\frac{0 - p(X)}{1 - p(X)} \cdot \Delta Y \;=\; -\frac{p(X)}{1 - p(X)} \cdot \Delta Y$$

Controls get reweighted by $p(X)/(1 - p(X))$ — and with a negative sign.

The reweighting only happens to the control group's first differences.
That's because it's the Y^0 that's missing, not the Y^1.

Take expectations: under conditional PT, the operator recovers the ATT

Treated piece ($D = 1$ rows contribute ΔY):

$$\mathbb{E}[\Delta Y \cdot \nVdash\{D{=}1\}] \;=\; \Pr(D{=}1) \cdot \mathbb{E}[\, Y^1_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1\,]$$

Control piece ($D = 0$ rows contribute $-\frac{p(X)}{1-p(X)}\Delta Y$). By iterated expectations:

$$-\mathbb{E}\left[\tfrac{p(X)}{1-p(X)}\,\Delta Y \cdot \nVdash\{D{=}0\}\right] \;=\; -\mathbb{E}\left[\tfrac{p(X)}{1-p(X)} \cdot (1 - p(X)) \cdot \mathbb{E}[\Delta Y \mid D{=}0, X]\right]$$
$$\;=\; -\mathbb{E}\big[p(X)\,\mathbb{E}[\Delta Y \mid D{=}0, X]\big]$$

Apply conditional PT: $\mathbb{E}[\Delta Y^0 \mid D{=}0, X] = \mathbb{E}[\Delta Y^0 \mid D{=}1, X]$, so the control piece becomes

$$-\,\mathbb{E}\big[p(X)\,\mathbb{E}[Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1, X]\big] \;=\; -\,\Pr(D{=}1) \cdot \mathbb{E}[\, Y^0_{\text{post}} - Y^0_{\text{pre}} \mid D{=}1\,]$$

Add the two pieces: Y^0_{pre} cancels, leaving

$$\Pr(D{=}1) \cdot \mathbb{E}[\, Y^1_{\text{post}} - Y^0_{\text{post}} \mid D{=}1\,] \;=\; \Pr(D{=}1) \cdot \text{ATT}$$

Normalize by $\Pr(D{=}1) \approx N_1/N$:

$$\widehat{\delta}_{\text{IPW}} \;=\; \frac{1}{N_1} \sum_i \frac{D_i - p(X_i)}{1 - p(X_i)}\,\Delta Y_i \;\to\; \text{ATT}. \quad \blacksquare$$

Common support is non-negotiable

- IPW's weights are $\dfrac{p(X)}{1-p(X)}$ for controls. As $p(X) \to 1$, the weight $\to \infty$.
- If some treated X-values have $p(X) \approx 1$, then NO control unit is comparable — and the weights are unstable.
- Standard responses:
  - Trim units with $p(X) > 0.95$ (or < 0.05).
  - Report sensitivity to the trim threshold.
  - In the limit, "weak overlap" breaks identification entirely. (Advanced deck.)

The pscore optimizer (IRLS, Newton-Raphson) can also misbehave near separation. Forward link to advanced.

Lesson

3. Doubly Robust DiD

Doubly robust: two chances to be right

- OR needs $\widehat{\mu}_0(X)$ correctly specified.
- IPW needs $\widehat{p}(X)$ correctly specified.
- Neither is foolproof. Each fails if its own model is wrong.

The Sant'Anna & Zhao (2020) move

Combine them. The **DR-DiD** estimator is consistent if *either* model is correct — not necessarily both.

Two chances to be right. That is what "doubly robust" means.

DR-DiD identification: three assumptions

Assumption 1 (Data)

Panel or repeated cross-sections with pre- and post-periods; treatment in post-period; covariates X pre-treatment.

Assumption 2 (Conditional parallel trends)

$$\mathbb{E}[\Delta Y(0) \mid D{=}1, X] \;=\; \mathbb{E}[\Delta Y(0) \mid D{=}0, X]$$

Assumption 3 (Common support)

$$0 \;<\; p(X) \;<\; 1 \text{ for almost every } X.$$

DR-DiD + these three $\Rightarrow$ ATT identified if either model is correct. Proof next slide.

The DR-DiD estimator

$$\widehat{\delta}_{\mathsf{DR}} = \mathbb{E}\left[\left(\frac{D}{\mathbb{E}[D]} - \frac{\frac{p(X)(1-D)}{1-p(X)}}{\mathbb{E}\left[\frac{p(X)(1-D)}{1-p(X)}\right]} \right) \left(\Delta Y - \mu_0(X)\right) \right]$$

- $p(X)$: propensity score model
- $\mu_0(X)$: outcome model fitted on *controls*: $\mu_0(X) = \mathbb{E}[\Delta Y \mid D{=}0, X]$
- The IPW weight reweights the comparison. The OR term centers the residual on the imputed counterfactual.
- If p is right, the IPW piece does the work. If μ_0 is right, the OR piece does the work. Proof next slide.

Proof: doubly robust — *either* model can be wrong

Let $w(X)$ denote the weight operator in $\widehat{\delta}_{\text{DR}} = \mathbb{E}[w(X)(\Delta Y - \mu_0(X))]$.

Case A: $p(X)$ **correct,** $\mu_0(X)$ **may be wrong.**

$$\widehat{\delta}_{\text{DR}} = \underbrace{\mathbb{E}[w(X)\,\Delta Y]}_{\text{IPW identifies ATT}} - \underbrace{\mathbb{E}[w(X)\,\mu_0(X)]}_{=0\text{ in expectation}} = \text{ATT.}$$

The OR term vanishes because $\mathbb{E}[w(X) \mid X] = 0$ once $p(X)$ is correct. **IPW alone identifies the ATT.**

Case B: $\mu_0(X)$ **correct,** $p(X)$ **may be wrong.** Under conditional PT, the residual $\Delta Y - \mu_0(X)$ has conditional mean zero given $D{=}0, X$:

$$\mathbb{E}[\Delta Y - \mu_0(X) \mid D{=}0, X] = 0,$$

so the control piece zeroes regardless of weighting. The treated piece $\frac{D}{\mathbb{E}[D]}(\Delta Y - \mu_0(X))$ integrates directly to the ATT. **OR alone identifies the ATT.**

Either suffices — not both. That is "two chances to be right." ■

DR is *not* regression-with-pscore

- Look at the formula. $\widehat{p}(X)$ **appears as a weight, never as a regressor.**
- $\mu_0(X)$ models the outcome, but on *controls only* — to impute the counterfactual trend.
- The two pieces are combined by a re-centering and reweighting, NOT by adding $\widehat{p}$ to a regression's right-hand side.

The pscore is the tool. ***Using it as a weight is doubly robust.*** *Using it as a regressor is the trap from Lesson 4.*

Efficiency and the influence function

- Under conditional PT + common support, DR-DiD is **semiparametrically efficient**.
- No estimator of the ATT that uses only the same assumptions can have a smaller asymptotic variance.
- The asymptotic variance has a closed form via the *influence function* (this is what the `drdid` package computes for SEs).

Practical recipe

Pick X for confounding (Lesson 1). Estimate $\widehat{p}(X)$ by logit. Estimate $\widehat{\mu}_0(X)$ on controls. Plug into the DR-DiD formula. Influence-function SEs.

The modern frontier: double-debiased ML

Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey & Robins (2018), *The Econometrics Journal*.

Same DR architecture — model the outcome, model the treatment, combine them. Two extensions:

- **Replace logit / OLS with any ML learner.** Lasso, random forests, gradient boosting, neural nets. The DR formula stays.
- **Cross-fitting.** Estimate $\widehat{p}$ and $\widehat{\mu}_0$ on one half of the sample, plug in on the other. Swap. Keeps the influence-function argument valid even when the nuisance models overfit.

Same lesson, more flexible tools. DR is what makes the ML extension principled — without doubly-robustness, fitting nuisance models with ML would invalidate the standard errors.

Reading: Chernozhukov et al. (2018, Econometrics Journal). The DRDID + ML combination is the active frontier.

DR-DiD, in code

```
library(DRDID)

# Panel data: outcome y, treatment indicator d, time t, id, covariates X
result <- drdid(
  yname = "y",
  tname = "t",
  idname = "id",
  dname = "d",
  xformla = ~ x1 + x2 + x3, # covariates for BOTH p(X) and mu_0(X)
  data = panel_df,
  estMethod = "imp", # Sant'Anna & Zhao (2020) DR
  weightsname = NULL,
  boot = FALSE,
  inffunc = TRUE # influence-function SEs
)
summary(result)
```

Same X used for both models. Misspecify one — the other catches you.

A Monte Carlo to settle the rankings

- Compare four estimators on the same data: **TWFE** (with X as additive controls), **OR**, **IPW**, **DR**.
- **4 DGPs** (Sant'Anna's design): vary whether the OR model and the pscore model are correctly specified.
- **Sample size:** 1,000 units. **Replications:** 10,000.
- **Report:** bias, RMSE, SE, 95% coverage, CI length.

The question we want answered: when does DR pay off, and when does it break?

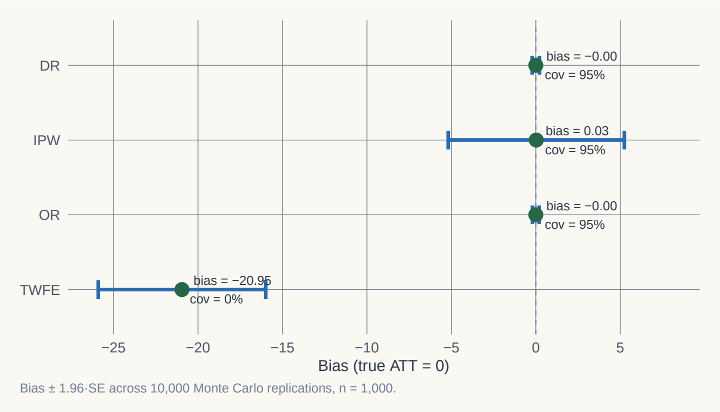
Driver code: `simulations/covariates/covariates_monte_carlo.{do,R}`

DGP 1 — both models correct

	Bias	RMSE	SE	Coverage	CI length
TWFE	-20.95	21.12	2.53	0.000	9.91
OR	-0.001	0.10	0.10	0.950	0.40
IPW	$+0.026$	2.77	2.66	0.952	10.44
DR	-0.001	0.11	0.11	0.947	0.41

*TWFE catastrophic (Bias ≈ -21; coverage zero). OR & DR both efficient (Bias ≈ 0, coverage 95%). IPW unbiased but **wide CIs**.*

DGP 1 — visualizing the sampling distribution



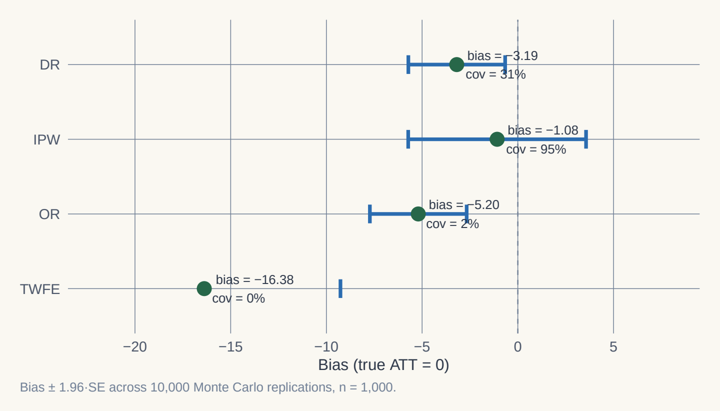
When both models are correct, OR and DR concentrate tightly on the truth. IPW is unbiased but noisy. TWFE is off the map.

DGP 4 — both models *wrong*

	Bias	RMSE	SE	Coverage	CI length
TWFE	-16.38	16.54	3.63	0.000	14.22
OR	-5.20	5.36	1.29	0.015	5.05
IPW	-1.08	2.66	2.37	0.949	9.31
DR	-3.19	3.45	1.29	0.308	5.07

DR is not magic. When both models fail, DR is biased and undercovers (0.31 vs. nominal 0.95).
IPW alone now wins on coverage — it doesn't rely on the outcome model.

DGP 4 — the sampling distribution tells the same story



DR no longer hugs the truth. IPW is wider but centered. Doubly robust ≠ doubly unbeatable.

What the Monte Carlo teaches

- **TWFE with additive X:** never wins. Coverage is zero in every DGP. **Stop using it as a default.**
- **OR:** efficient when the outcome model is right. Disaster when it's wrong (Bias -5, coverage 1.5%).
- **IPW:** less efficient but *robust to outcome misspecification*. The unsexy workhorse.
- **DR:** best of both *when at least one is right*. Not better than IPW when *both* are wrong.

"Doubly robust" means robust to one misspecification. Not two. Not magic.

Replication: `simulations/covariates/covariates_monte_carlo.{do,R}`; figures `mc_dr_1.png`, `mc_dr_2.png`.

Closing addendum: DR is *not* regression-plus-pscore

A common misdescription you'll hear:

> *Doubly robust DiD is just outcome regression plus the propensity score as another control.*

— Heard at more than one seminar

No. That description conflates three different things:

- what **outcome regression** actually does (model $\mu_0(X)$ on controls; impute counterfactuals),
- what **IPW** actually does (use $\widehat{p}(X)$ as a *weight*, not a regressor),
- what **DR** actually does (combine the two).

Adding $\widehat{p}(X)$ as a regressor on the right-hand side is its own — different — estimator. It's a trap. The next few slides walk through exactly why.

A natural temptation

> *If the propensity score is the one-scalar summary of confounding, why not just add it as a regressor? Done.*

— The audience, right now

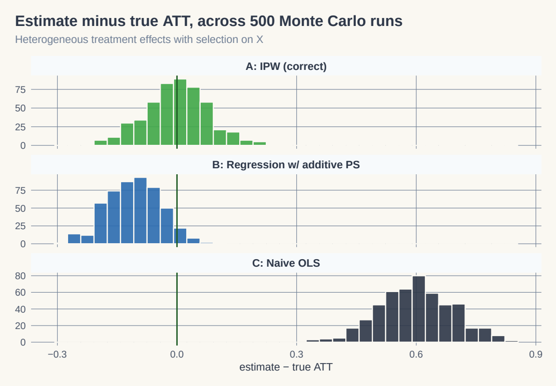
- It sounds right. Rosenbaum-Rubin says *one scalar suffices*. Regression handles scalars.
- So: $Y_i = \alpha + \delta D_i + \gamma \widehat{p}(X_i) + \varepsilon_i$
- Surely this recovers δ as the ATT? **No.** And the reason matters.

What that regression actually assumes

$$Y_i = \alpha + \delta D_i + \gamma\,\widehat{p}(X_i) + \varepsilon_i$$

- No $D \times X$ interaction. No $D \times \widehat{p}$ interaction.
  The treatment effect is forced constant across the entire range of $\widehat{p}(X)$.
- Equivalently: the same dose of policy produces the same effect on a person with $\widehat{p}(X) = 0.10$ as on a person with $\widehat{p}(X) = 0.90$.
- That's an assumption about **the outcome side** — one that Rosenbaum-Rubin says nothing about.
- Rosenbaum-Rubin gives you a valid *reweighting* tool. It does not authorize regression-on-pscore.

A simulation makes it visible



*500 reps. Heterogeneous $\delta(X) = 1 + 0.5\,X_1$. Method A (IPW reweighting): bias ≈ 0. Method B (regression with $\widehat{p}$): **bias** $\approx -10\%$.*

Code: `glasgow/decks/code/IPW-Demo/ipw_simulation.R`.

The simulation, one screen of R

```
N <- 1000
X1 <- rnorm(N); X2 <- rnorm(N)
p <- plogis(0.5*X1 + 0.3*X2 - 0.2) # true propensity
D <- rbinom(N, 1, p)
Y0 <- X1 + 0.5*X2 + rnorm(N)
Y1 <- Y0 + (1 + 0.5*X1) # heterogeneous tau(X)
Y <- D*Y1 + (1-D)*Y0

phat <- glm(D ~ X1 + X2, family=binomial)$fitted.values

# Method A: IPW -- reweight controls
w <- ifelse(D==1, 1, phat/(1-phat))
A <- mean(Y[D==1]) - weighted.mean(Y[D==0], w[D==0])

# Method B: regression with pscore as a control
B <- coef(lm(Y ~ D + phat))["D"] # <-- the trap
```

Full simulation: `decks/code/IPW-Demo/ipw_simulation.R`

Weight versus regressor — different uses of the same scalar

- The propensity score is a **tool**. The same scalar can be used as:
  - a **weight** (Method A: reweight controls) — recovers ATT
  - a **regressor** (Method B: add as control) — imposes homogeneous effects
- Same number. Different uses. *Different identifying assumptions.*
- Method B's bias is not noise. It is the heterogeneity in $\delta(X)$ that the regression cannot see.

The lesson before DR

"Doubly robust" is sometimes mis-summarized as "IPW plus a regression control."

That's wrong. DR is something else — it uses the pscore as a weight, AND it uses an outcome model.
But it does NOT use the pscore as a regressor.

Name the disease: ***additive-control-side bias***

Whenever you put a covariate on the right-hand side of an outcome regression *additively* (no $D \times X$ interaction), you force:

$$\delta \text{ is constant across all values of that covariate.}$$

- Method B (Lesson 4): the covariate is $\widehat{p}(X)$ — a scalar summary of X.
  Force constant δ across $\widehat{p}(X)$. Heterogeneity in $\delta(X)$ is invisible.
- TWFE-with-controls (Lesson 2): the covariate is X itself.
  Force constant δ across X. Heterogeneity in $\delta(X)$ is invisible.
- **Same disease, different costume.** Caetano & Callaway (2024) name the TWFE version *hidden linearity bias*. It's the same root.

DR's escape: use X and $\widehat{p}(X)$ as weights and to model the counterfactual $\mu_0(X)$ on controls only — never as additive regressors of Y.

Lesson

4. Hidden Linearity Bias

Caetano & Callaway 2024

TWFE-with-controls has two distinct problems

The workhorse specification $Y_{it} = \alpha_i + \gamma_t + \delta D_{it} + X_{it}'\beta + \varepsilon_{it}$ is biased for two *separate* reasons:

1. **The FE transformation throws away what you wanted to control for.**
   First-differencing X_{it} leaves only ΔX. The *level* of X is gone. Time-invariant Z is gone entirely. (Caetano-Callaway 2024 — this lesson.)
2. **What's left enters additively, forcing constant δ.**
   Whatever covariate survives the FE transformation sits on the RHS as an additive regressor with no $D \times X$ interaction. Same disease as Method B (Lesson 4).

This lesson covers problem (1). Problem (2) is the L4 callback we'll close on.

An applied problem you've already run

> *TWFE with time-varying controls is the workhorse DiD specification. Almost every DiD paper runs it.*

— Anybody who's seen a job-market paper

- Standard recipe: include population, income, demographic shares, region indicators, etc. on the right-hand side. Run TWFE. Done.
- **What if I told you that regression doesn't control for what you think it controls for?**
- Caetano & Callaway (2024, R&R *REStat*): yes. The fixed-effect transformation quietly drops your X levels and your time-invariant Z.

An application: stand-your-ground laws (Cheng & Hoekstra 2013)

- 50 states, 2000–2010. 20 states adopt stand-your-ground laws. Outcome: log homicide rate.
- Treated states differ from controls in **levels**:
  - South dummy: std. diff. $= 0.81$ (large)
  - Poverty rate: std. diff. $= 1.06$ (huge)
  - Median income: std. diff. $= -1.08$ (huge)
- But the differences in **changes** of these covariates are small.

The headline result

TWFE with covariates: $\widehat{\alpha} = +0.067$, significant.
AIPW with the right covariates: $\widehat{\text{ATT}} \approx 0$, insignificant.

The positive TWFE result is largely artefact of hidden linearity bias.

What TWFE actually controls for, after the FE transformation

You wrote down this model:

$$Y_{it} \;=\; \alpha_i \;+\; \gamma_t \;+\; \delta D_{it} \;+\; X'_{it}\beta \;+\; Z'_i\eta \;+\; \varepsilon_{it}$$

First-difference (or within-transform) to kill α_i:

$$\Delta Y_{it} \;=\; \widetilde{\gamma}_t \;+\; \delta\,\Delta D_{it} \;+\; \Delta X'_{it}\beta \;+\; \Delta\varepsilon_{it}$$

- α_i vanishes (good).
- Z_i vanishes too (time-invariant). **Region: gone.**
- X_{it} becomes ΔX_{it}. **Population level: gone. Only population growth remains.**
- Your "controls" are now *changes* in X, not levels.

What you should be controlling for

The correct first-differenced model for $\Delta Y(0)$ (Caetano-Callaway Eq. 4):

$$\Delta Y_{it}(0) = \tilde{\gamma}_t + \underbrace{Z_i' \tilde{\delta}_t}_{\text{time-inv}} + \underbrace{\Delta X_{it}' \beta_t}_{\text{changes}} + \underbrace{X_{i,t-1}' \tilde{\beta}_t}_{\text{level}} + \Delta\varepsilon_{it}$$

- **Time-invariant** Z (region) belongs on the right — because trends differ by region.
- **Pre-period level** $X_{i,t-1}$ belongs on the right — because the level of population predicts the trend in homicides.
- TWFE drops both. **Hidden linearity bias.**

The L4 callback: problem (2) is the additive-RHS disease

We've covered problem (1) above — the FE transformation throws away levels.
Problem (2) is the L4 disease, present in TWFE-with-controls too.

Estimator	What goes on RHS additively	What gets forced constant
Method B (L4)	$\widehat{p}(X)$ as control	δ across $\widehat{p}(X)$
TWFE-with-controls (L2, L6)	ΔX as control	δ across ΔX
General formulation	$W = f(X)$ as additive RHS regressor	δ across W

- Problem (1) is C&C's contribution: *what* you control for is wrong.
- Problem (2) is the L4 disease: *how* you control for it is wrong.
- TWFE-with-controls catches both. DR escapes both — using X and $\widehat{p}(X)$ as **weights**, not as additive regressors.

The fix: AIPW with $(X_{t-1},\ Z)$

Caetano-Callaway's recommended estimator:

$$\widehat{\text{ATT}} = \widehat{\mathbb{E}}_{D=1}[\Delta Y - \widehat{m}_0(X_{t-1}, Z)]$$

- $\widehat{m}_0(X_{t-1}, Z)$: outcome regression of ΔY on *pre-treatment level* of X and *time-invariant* Z — fit on **controls only**.
- **Why X_{t-1}, not X_t?** Because X_t in the post-period could itself be affected by treatment. The pre-treatment level is uncontaminated.
- Combine with IPW on a generalized propensity score (same conditioning set) for full doubly-robust safety.
- CC's fully preferred spec also includes $\Delta X = X_t - X_{t-1}$ for dimension reduction. R: `pte`, `twfeweights`.

Pre-treatment X is uncontaminated — post-treatment X is not

> *Why X_{t-1} and not X_t? You're throwing away the most recent data point.*

— A student, raising a hand

- X_t sits **downstream of treatment**. If the policy moves X — even a little — you've controlled for a *post-treatment outcome*.
- Conditioning on a post-treatment variable doesn't reduce bias. It **re-introduces** bias — the bad-controls problem (Angrist–Pischke 2009, Ch. 3).
- X_{t-1} is settled before D is assigned. It carries the selection information without carrying any of the treatment.

The rule generalizes: condition on the latest pre-treatment value, never the contemporaneous one.

Caetano & Callaway (2024), §3; classic "bad controls" point goes back to Rosenbaum (1984).

Stand-your-ground revisited — the numbers

Specification	$\widehat{\text{ATT}}$	**SE**
TWFE with ΔX (the standard recipe)	$+0.067$	0.026
Reg. adj. with $\Delta X + Z$	$+0.106$	0.058
Reg. adj. with $X_{t-1} + Z$ (*Caetano-Callaway preferred*)	$+0.019$	0.043
Reg. adj. with $\Delta X + X_{t-1} + Z$	$+0.078$	0.180

- The TWFE positive effect ($+6.7\%$) shrinks to near-zero ($+1.9\%$) when you control for *levels* of X and Z.
- The bias was hidden in plain sight — the regression compiled fine. It just wasn't controlling for what we thought.

Lesson

5. Choosing & checking X

The practical close — picking, balance, and when controls are unnecessary

Picking X: theory or LASSO on $\Delta Y(0)$

You want X that (i) predicts the treatment *and* (ii) predicts the untreated trend.
Pre-treatment only — post-treatment X can be affected by D and will leak.

Two paths to choose which X:

- **Theory / common sense.** Variables you believe drive both selection and the untreated trend. This is how most of it actually gets done.
- **Data-driven.** LASSO (or another selector) on

$$\Delta Y(0)_i \;\sim\; X_i, \quad \text{fit on untreated unit-periods only.}$$

  Keep what LASSO keeps.

The regressand is $\boldsymbol{\Delta Y(0)}$. Not $Y(1)$. Not post-treatment Y — that's $Y(1)$, and using it lets D leak through the back door. This is **Heckman-Ichimura-Todd (1997)** logic: the data can only teach you which X's predict the untreated counterfactual trend.

Beyond that — functional form, interactions, transformations — the data is mostly silent.

If X is balanced, you don't need to control for it

Same logic as an RCT: balanced X across treatment and control makes the comparison fair without conditioning.

Worked example. Let sex be the only confounder. Within sex, the untreated trends are:

$$\Delta Y^0_{\text{men}} = 10 - 6 = 4 \qquad \Delta Y^0_{\text{women}} = 8 - 5 = 3.$$

Balanced — both groups 50% male.

$$\mathbb{E}[\Delta Y^0 \mid D{=}1] = 0.5(4) + 0.5(3) = 3.5 = \mathbb{E}[\Delta Y^0 \mid D{=}0].$$

Unconditional PT holds. You didn't condition on sex — and you didn't need to.

Imbalance turns a confounder into bias

Same untreated trends as before — men $+4$, women $+3$ — but now the groups have different sex composition:

Imbalanced — treated 75% male, control 25% male.

$$\begin{aligned} \mathbb{E}[\Delta Y^0 \mid D{=}1] &= 0.75(4) + 0.25(3) = 3.75 \\ \mathbb{E}[\Delta Y^0 \mid D{=}0] &= 0.25(4) + 0.75(3) = 3.25. \end{aligned}$$

Gap $=$ 0.5. That's the bias a naive DiD on ΔY would carry.

Takeaway. Balance is what licenses unconditional PT. Imbalance turns a Y^0-predictor into a bias on ΔY^0.

Check the balance: normalized differences

Before you trust your X, check that the treated and control groups are *actually* comparable on it.

Normalized difference

$$\text{Norm. Diff}_X = \frac{\overline{X}_T - \overline{X}_C}{\sqrt{(S_T^2 + S_C^2)/2}}$$

- A unit-free imbalance measure — comparable across variables.
- **Rule of thumb: < 0.25.** Above that, classical OLS/IPW have trouble.
- Unlike a t-statistic, it doesn't shrink to zero just because the sample is small.

Imbens & Rubin (2015), Causal Inference for Statistics, Social, and Biomedical Sciences, Ch. 14.

What balance tells you (and what it doesn't)

A confounder has *two* jobs:

1. $X \to D$: different distribution in treated vs. control.
2. $X \to \Delta\mathbb{E}[Y^0]$: predicts the untreated outcome trend.

The balance check only diagnoses (1).

- Pick X theoretically — variables you believe predict Y^0 trends.
- *Then* run normalized differences to see whether they're differentially distributed.
- Imbalance + predictiveness $\Rightarrow$ controlling matters. Imbalance without predictiveness $\Rightarrow$ noise.

Don't over-spend on already-balanced X

Adding controls in OLS DiD is cheap. In IPW or DR, it isn't.

The propensity score is fit by logistic regression, which is fragile in high dimensions:

Events-per-variable rule

Roughly **5–10 events per covariate** for stable logistic estimates,
where "events" $=$ treated units.

Implication. A balanced X that doesn't strongly predict ΔY^0 doesn't deserve a slot in the pscore model. The balance table tells you where to spend the budget. *EPV guideline: Peduzzi et al. (1996), J. Clin. Epidemiol.*

A balance table in the wild

Table 4: Covariate Balance Statistics

	Unweighted			Weighted		
Variable	Non-Adopt	Adopt	Norm. Diff.	Non-Adopt	Adopt	Norm. Diff.
2013 Covariate Levels						
% Female	27.94	28.32	0.19	29.89	30.13	0.14
% White	47.43	52.50	0.62	46.07	47.76	0.21
% Hispanic	5.86	4.75	-0.14	10.06	11.35	0.13
Unemployment Rate	7.10	7.77	0.25	6.98	8.00	0.50
Poverty Rate	18.54	16.22	-0.35	17.22	15.29	-0.37
Median Income	43.38	48.00	0.41	49.28	57.83	0.68
2014 - 2013 Covariate Differences						
% Female	-0.11	-0.13	-0.06	-0.05	-0.04	0.10
% White	-0.29	-0.35	-0.14	-0.29	-0.27	0.07
% Hispanic	0.11	0.11	-0.01	0.14	0.20	0.33
Unemployment Rate	-1.09	-1.26	-0.24	-1.08	-1.36	-0.54
Poverty Rate	-0.51	-0.28	0.13	-0.41	-0.35	0.04
Median Income	1.13	1.04	-0.04	1.11	1.73	0.32

This table reports the covariate balance between adopting and non-adopting states. In the top panel, we report the averages and standardized differences of each variable, measured in 2013, by adoption status. In the bottom panel we report the average and standardized differences of the county-level long differences between 2014 and 2013 of each variable. We report both weighted and unweighted measures of the averages to correspond to the different estimation methods of including covariates in a 2 × 2 setting.

Every variable, its treated mean, its control mean, the normalized difference. One column tells you whether the design will make you sweat.

What we’ve done this afternoon

1. Conditional PT — the new identifying assumption
2. Outcome regression: model the trend, impute the counterfactual
3. IPW: reweight controls to look like the treated
4. Doubly robust: pscore as a *weight* (not regressor) + outcome model — and why *controlling for* the pscore is the trap
5. Hidden linearity bias (Caetano-Callaway): *same disease*, costume = TWFE-with-controls

The unifying lesson: covariates on the additive RHS of an outcome regression impose constraints on δ.
*DR’s escape is to use them as **weights**, not regressors.*

Tomorrow

- **Day 3:** continuous treatment — when the dose itself varies.
- Plus: when treatment timing differs across units — the staggered-adoption mess (Goodman-Bacon, Callaway-Sant'Anna, Sun-Abraham, BJS, dCdH).

Use the pscore as a **weight,**

not as a **regressor.**

That is the difference between **doubly robust** and **doubly biased.**